

Clinical Effectiveness

METIMMOX-1 Economic Evaluation

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1 Overview

This report presents clinical effectiveness results from the METIMMOX-1 trial, evaluating biomarker-guided treatment strategies for selecting immunotherapy candidates in metastatic MSS/pMMR colorectal cancer.

The analysis uses **Firth-corrected Cox proportional hazards regression** with profile likelihood confidence intervals to assess treatment effect heterogeneity across two pre-treatment biomarkers identified by the causal DAG analysis:

- **CRP** (C-reactive protein): Low CRP (<5 mg/L) at baseline — a pre-treatment prognostic and potentially predictive marker
- **TMB/BRAF**: High tumor mutational burden (>9 mut/MB) or BRAF mutation — a pre-treatment genomic marker

Model specification is motivated by the causal DAG (see `dag.qmd` and `dag_associations.qmd`).

The DAG encodes treatment (T) as randomized with no parents, so no confounding adjustment is needed for causal identification; Age and Sex are included as precision covariates. TLR (Tumor Lesion Reduction) is encoded as a post-randomization intermediate variable ($T \rightarrow \text{TLR} \rightarrow \text{PFS}$): it is not a pre-treatment baseline characteristic and conditioning on it as a covariate would block part of the treatment effect pathway. TLR is therefore excluded from the primary survival models and examined separately as a descriptive endpoint.

The **unified DAG-informed model** is:

$$\text{Surv}(\text{time}, \text{event}) \sim \text{Age} + \text{sex} + \text{Rx} + \text{CRP} + \text{TMB/BRAF} + \text{CRP} \times \text{Rx} + \text{TMB/BRAF} \times \text{Rx}$$

Ridge regression (L2-penalized Cox) is applied as a sensitivity analysis to assess the stability of the interaction estimates under shrinkage. TLR is reported in a separate exploratory section.

2 Methodological Notes

2.1 Software Packages

This analysis uses the following R packages:

- `coxphf` version 1.13.4: Firth-corrected Cox regression with profile likelihood CIs
- `survival` version 3.7.0: Cox regression and Schoenfeld residual diagnostics
- `glmnet` version 4.1.8: Ridge regression sensitivity analysis
- `vcd` version 1.4.13: Cramér's V for biomarker correlation assessment

2.2 Firth-Corrected Cox Regression

We use Firth's penalized maximum likelihood method for Cox regression, which addresses:

1. **Monotone likelihood**: When a covariate perfectly separates events from non-events
2. **Small sample bias**: Reduces bias in coefficient estimates with sparse data
3. **Unstable confidence intervals**: Profile likelihood CIs are more reliable than Wald-based CIs in small samples

Statistical inference is based on **penalized likelihood ratio tests (PLRT)**, which are more appropriate than Wald tests for testing treatment-biomarker interactions in this setting.

2.3 Ridge Regression Sensitivity Analysis

We use Ridge regression as a sensitivity analysis to assess the stability of our interaction estimates. Ridge regression (also called L2-penalized regression or Tikhonov regularization) belongs to the family of regularized or penalized regression methods. In survival analysis, this is implemented as a penalized Cox model.

How Ridge regression works:

- Ridge adds a penalty term that discourages large coefficients. The bigger a coefficient tries to get, the more the model resists, pulling estimates toward zero (HR toward 1.0).
- Unlike LASSO ($\alpha = 1$), which sets some coefficients exactly to zero and performs variable selection, Ridge ($\alpha = 0$) shrinks all coefficients but retains all predictors.
- The degree of shrinkage is controlled by the tuning parameter λ , which is selected via **cross-validation (CV)**: the data are repeatedly split into training and validation sets, and the λ that produces the best predictive performance is chosen.

Why not LASSO or Elastic Net?

We determined that LASSO and elastic net are **not appropriate** for this analysis:

1. **Correlated biomarkers:** LASSO arbitrarily selects one of correlated predictors while setting others to zero, which does not reflect the clinical question of comparing ALL biomarker strategies.
2. **Hierarchy principle violation:** LASSO may select interaction terms (e.g., `crp:Rx`) while setting the main effect (`crp`) to zero, producing models that are difficult to interpret clinically.
3. **Clinical objective:** Our goal is to estimate treatment effect heterogeneity across ALL biomarkers simultaneously, not to select a “winning” biomarker.

Elastic net ($\alpha = 0.1$ – 0.3) represents a compromise but does not fully address these concerns.

Interpreting Ridge results:

Ridge regression is not replacing the Firth estimates—it provides a reality check. When Ridge substantially shrinks an interaction coefficient toward zero while Firth estimates a large effect, this indicates the effect is too unstable to trust for practical use. The CV-selected λ reflects how much regularization is needed for reasonable out-of-sample prediction given the sample size and collinearity structure.

3 Data Preparation

3.1 Sample Characteristics

3.2 Biomarker Correlations

Note: Cramér’s V interpretation: <0.1 = Negligible, 0.1 – 0.3 = Weak, >0.3 = Moderate/Strong. The observed correlations are weak to negligible. While modest, any correlation among candidate biomarkers supports our decision not to use LASSO, which would arbitrarily select among correlated predictors.

Table 1: Sample Characteristics by Biomarker Status

Characteristic	Overall	CRP		TLR		TMB/BRAF	
		Positive	Negative	Positive	Negative	Positive	Negative
Patients, n	65	22	43	41	24	29	36
Demographics							
Age, mean (SD)	64.0 (10.0)	64.3 (10.3)	63.8 (10.0)	65.0 (9.9)	62.2 (10.2)	64.6 (10.4)	63.5 (9.8)
Male	30 (46.2%)	12 (54.5%)	18 (41.9%)	19 (46.3%)	11 (45.8%)	14 (48.3%)	16 (44.4%)
Female	35 (53.8%)	10 (45.5%)	25 (58.1%)	22 (53.7%)	13 (54.2%)	15 (51.7%)	20 (55.6%)
Treatment arm							
Control (FLOX alone)	29 (44.6%)	5 (22.7%)	24 (55.8%)	22 (53.7%)	7 (29.2%)	13 (44.8%)	16 (44.4%)
Experimental (FLOX/nivolumab)	36 (55.4%)	17 (77.3%)	19 (44.2%)	19 (46.3%)	17 (70.8%)	16 (55.2%)	20 (55.6%)
Outcome events							
Deaths (OS events)	56 (86.2%)	16 (72.7%)	40 (93.0%)	32 (78.0%)	24 (100.0%)	22 (75.9%)	34 (94.4%)
Progressions (PFS events)	48 (73.8%)	14 (63.6%)	34 (79.1%)	27 (65.9%)	21 (87.5%)	19 (65.5%)	29 (80.6%)
Biomarker prevalence							
CRP-positive (low CRP)	22 (33.8%)	22 (100.0%)	0 (0.0%)	18 (43.9%)	4 (16.7%)	11 (37.9%)	11 (30.6%)
TLR-positive (early response)	41 (63.1%)	18 (81.8%)	23 (53.5%)	41 (100.0%)	0 (0.0%)	18 (62.1%)	23 (63.9%)
TMB/BRAF-positive	29 (44.6%)	11 (50.0%)	18 (41.9%)	18 (43.9%)	11 (45.8%)	29 (100.0%)	0 (0.0%)

Note: Percentages within each column use the column subgroup as denominator. CRP/TLR/TMB-BRAF rows in the corresponding subgroup column equal 100% by definition.

Table 2: Biomarker Correlations (Cramér’s V)

Biomarker Pair	Cramér’s V	Interpretation
CRP vs TLR	0.278	Weak
CRP vs TMB/BRAF	0.077	Negligible
TLR vs TMB/BRAF	0.019	Negligible

4 Proportional Hazards Assumption

4.1 Schoenfeld Residual Tests

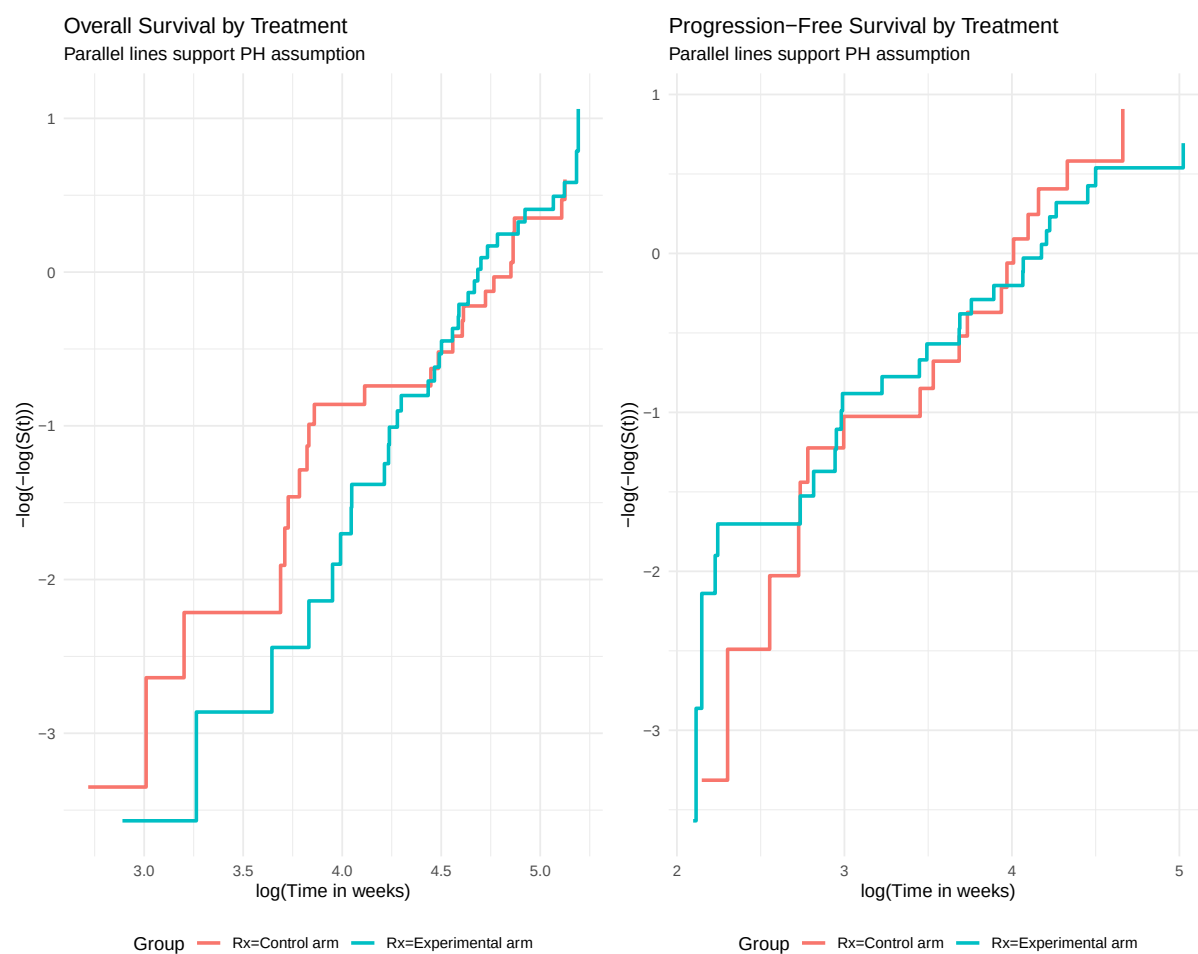
The proportional hazards assumption was tested using scaled Schoenfeld residuals. A significant p-value indicates potential violation of the PH assumption for that covariate.

Table 3: Schoenfeld Residual Test for Proportional Hazards Assumption

Covariate	χ^2 (OS)	p-value	χ^2 (PFS)	p-value
Age	1.85	0.174	0.50	0.480
sex	0.10	0.751	0.04	0.847
Rx	3.15	0.076	0.04	0.841
crp_num	0.89	0.345	3.22	0.073
tmb_braf_num	1.01	0.315	5.33	0.021
Rx:crp_num	2.26	0.133	1.21	0.272
Rx:tmb_braf_num	0.57	0.450	4.01	0.045
GLOBAL	7.32	0.396	19.60	0.007

Note: GLOBAL is the omnibus test for all covariates. Individual covariate tests help identify specific PH violations.

4.2 Log-Log Survival Plots



Interpretation: Approximately parallel lines on the log-log plot support the proportional hazards assumption. Crossing or converging lines suggest time-varying effects.

5 Primary Analysis: Firth-Corrected Cox Models

5.1 Unified Model Results

The unified DAG-informed model includes CRP and TMB/BRAF as pre-treatment biomarkers with their treatment interactions, adjusted for Age and Sex.

Formula: $\text{Surv}(\text{time}, \text{event}) \sim \text{Age} + \text{sex} + \text{Rx} + \text{CRP} + \text{TMB/BRAF} + \text{CRP:Rx} + \text{TMB/BRAF:Rx}$

5.1.1 Overall Survival

Table 4: Unified Model: Overall Survival - Full Coefficient Table

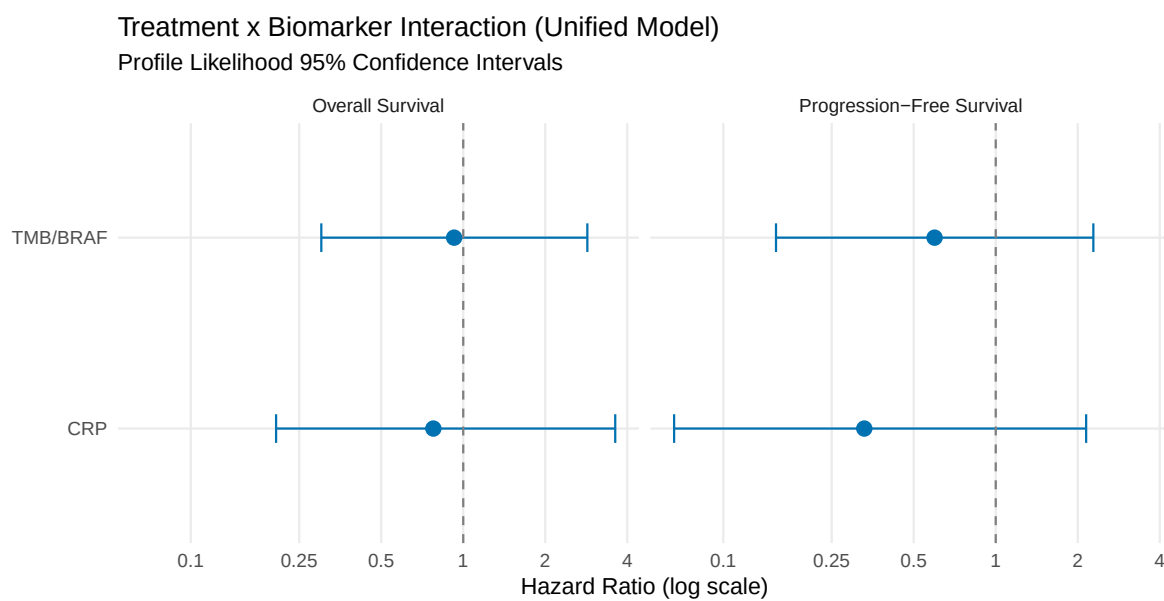
Term	HR (95% CI)	PLRT p-value
Treatment x Biomarker Interactions		
RxExperimental arm:crp_num	0.78 (0.21-3.61)	1.000
RxExperimental arm:tmb_braf_num	0.93 (0.30-2.85)	1.000
Main Effects and Covariates		
Age	1.00 (0.97-1.03)	—
sex1	1.58 (0.90-2.79)	—
RxExperimental arm	1.70 (0.81-3.67)	—
crp_num	0.52 (0.13-1.47)	—
tmb_braf_num	0.94 (0.41-2.12)	—

5.1.2 Progression-Free Survival

Table 5: Unified Model: Progression-Free Survival - Full Coefficient Table

Term	HR (95% CI)	PLRT p-value
Treatment x Biomarker Interactions		
RxExperimental arm:crp_num	0.33 (0.07-2.15)	1.000
RxExperimental arm:tmb_braf_num	0.60 (0.16-2.28)	1.000
Main Effects and Covariates		
Age	0.99 (0.96-1.02)	—
sex1	1.20 (0.65-2.26)	—
RxExperimental arm	2.01 (0.89-4.84)	—
crp_num	1.00 (0.19-3.60)	—
tmb_braf_num	0.77 (0.28-2.01)	—

5.2 Forest Plot



Interpretation: $HR < 1$ indicates that biomarker-positive patients derive greater benefit from experimental treatment (reduced hazard of death/progression). The dashed line at $HR = 1$ represents no differential treatment effect.

6 Sensitivity Analysis: Ridge Regression

Ridge regression (L2-penalized Cox, $\alpha = 0$) shrinks all coefficients toward zero but does not eliminate any, providing a sensitivity check for the stability of the interaction estimates. The same unified model formula is used; only the main effects (CRP and TMB/BRAF) are penalized.

Table 6: Overall Survival: Firth vs Ridge Regression

Interaction	Firth (95% CI)	Ridge (point est.)
CRP x Rx	0.78 (0.21-3.61)	0.39 (SE: 2.14)
TMB/BRAF x Rx	0.93 (0.30-2.85)	0.87 (SE: 2.04)

Note: Ridge: L2 shrinkage on main effects. Substantial shrinkage vs Firth = estimate sensitive to regularization.

Table 7: Progression-Free Survival: Firth vs Ridge Regression

Interaction	Firth (95% CI)	Ridge (point est.)
CRP x Rx	0.33 (0.07-2.15)	0.32 (SE: 2.15)
TMB/BRAF x Rx	0.60 (0.16-2.28)	0.45 (SE: 2.05)

^a coxphf convergence not attained for TMB/BRAF x Rx; interpret with caution.

^b Ridge applies L2 shrinkage to main effects only.

Interpretation: Firth provides the best unbiased point estimate given the sample size and data structure. Ridge applies L2 shrinkage toward zero (HR toward 1.0). Comparing the two methods reveals estimate stability: if Firth and Ridge agree closely, the estimate is robust; if Ridge substantially shrinks the interaction toward null, the effect is sensitive to regularization in this sample.

7 TLR as a Post-Treatment Response Measure

Causal DAG status: The DAG encodes TLR (Tumor Lesion Reduction) as a **post-randomization intermediate** ($T \rightarrow \text{TLR} \rightarrow \text{PFS}$). This means:

1. TLR is measured after treatment has started (requires at least one CT imaging cycle), so it cannot inform pre-treatment patient selection.
2. Including TLR as a covariate in a survival model alongside treatment would condition on a post-randomization variable, blocking part of the treatment-effect pathway (collider/mediator bias).
3. The strong $\text{TLR} \rightarrow \text{PFS}$ association observed in the DAG association tests (HR 0.18, $p < 0.001$) reflects a tautological relationship between radiological response and progression — tumors that respond to treatment have longer PFS by definition.

TLR is therefore reported here as a **descriptive endpoint only**, not as a baseline predictive biomarker. Its potential use as a clinical decision tool (e.g., stopping treatment in non-responders) would require a separate causal mediation analysis, which is beyond the scope of this report.

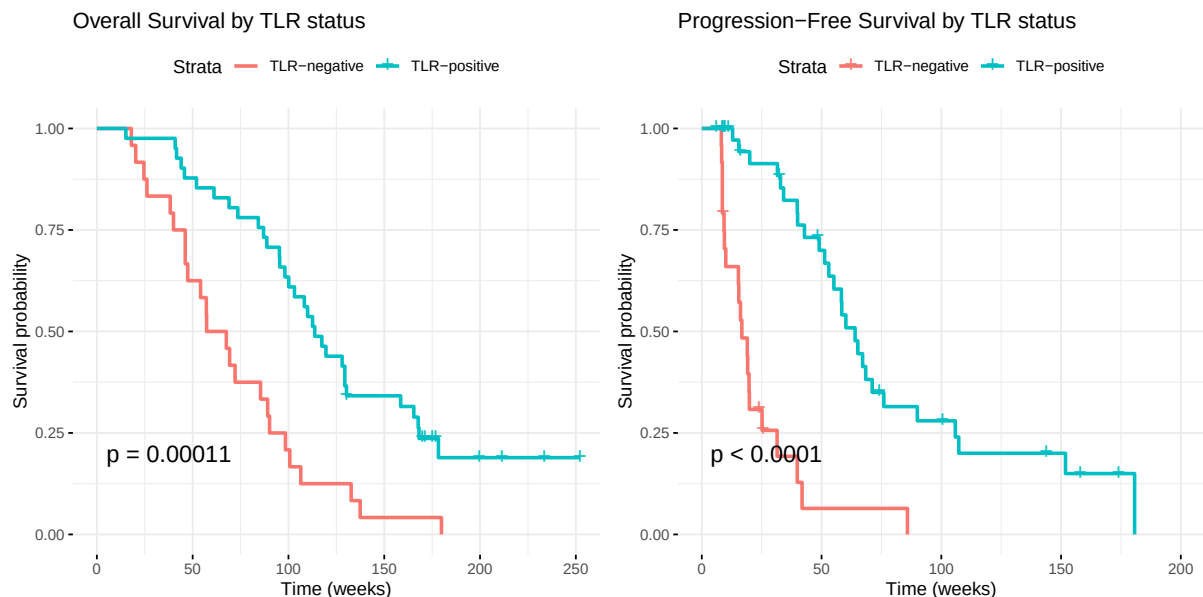
7.1 TLR Prevalence by Treatment Arm

Table 8: TLR Prevalence by Treatment Arm (Descriptive)

Treatment Arm	N	TLR-positive	%
Control (FLOX)	29	22	75.9
Experimental (FLOX/nivo)	36	19	52.8

Note: TLR = Tumour Lesion Reduction $\geq 10\%$ at first CT scan (post-treatment).

7.2 Kaplan-Meier Curves Stratified by TLR



Note: These curves are purely descriptive and stratify by a post-treatment response variable. The observed difference in survival by TLR status reflects treatment response, not a pre-treatment patient characteristic.

7.3 Unadjusted Prognostic Association (Descriptive)

Table 9: Unadjusted Prognostic Association of TLR (Descriptive Only)

Outcome	HR (95% CI)
Overall Survival	0.35 (0.20-0.61)
Progression-Free Survival	0.17 (0.09-0.32)
<i>Note:</i> TLR-pos. vs TLR-neg. Unadjusted Cox; TLR is a post-treatment mediator (not a causal estimate).	

8 Discussion

8.1 Primary interaction results

Neither biomarker-treatment interaction reached conventional significance in the unified model. For **overall survival**, $\text{CRP} \times \text{Rx}$ yielded HR 0.78 (95% CI 0.21–3.61, PLRT $p = 0.73$) and $\text{TMB/BRAF} \times \text{Rx}$ HR 0.93 (0.30–2.85, $p = 0.89$) — point estimates close to null with very wide confidence intervals. For **progression-free survival**, the $\text{CRP} \times \text{Rx}$ interaction showed a more directionally suggestive effect (HR 0.33, 0.07–2.15, $p = 0.22$): CRP-positive patients in the experimental arm had roughly one-third the hazard of progression relative to CRP-positive controls, though the CI spans more than an order of magnitude and the coxphf algorithm reported convergence difficulties for the $\text{TMB/BRAF} \times \text{Rx}$ PFS term (see footnote to Table 5). These results are consistent with the trial being underpowered to detect treatment-effect heterogeneity ($n = 65$, 48 progressions, 56 deaths): even a moderate subgroup effect (HR ~ 0.5) would require far larger samples for reliable estimation.

The PFS pattern is worth noting as a hypothesis-generating finding. The CRP direction (HR 0.33) is consistent with the DAG association test showing $\text{CRP} \rightarrow \text{PFS}$ (HR 0.40, $p = 0.004$) and the biological hypothesis that low CRP (reflecting lower systemic inflammation) identifies patients more likely to respond to immunotherapy. However, the OS CRP interaction estimate (HR 0.78) is considerably closer to null, suggesting that any PFS benefit does not clearly translate to an OS benefit in this dataset.

8.2 Proportional hazards and TMB/BRAF

The PH assumption was well supported in overall survival (Schoenfeld global $p = 0.40$). In progression-free survival, however, the global test was significant ($p = 0.007$), driven mainly by TMB/BRAF — both the main effect ($p = 0.021$) and the $\text{TMB/BRAF} \times \text{Rx}$ interaction ($p = 0.045$) violated the PH assumption. This indicates that the hazard ratio for TMB/BRAF (and its modification by treatment) is not constant over time in PFS, which may reflect different kinetics of early vs late events in TMB/BRAF-positive tumours. The PFS $\text{TMB/BRAF} \times \text{Rx}$ interaction estimate (HR 0.60) should therefore be interpreted with caution; a time-varying Cox model or landmark analysis would be more appropriate for that specific question.

8.3 Ridge regression sensitivity

For OS, Ridge shrinks the $\text{CRP} \times \text{Rx}$ estimate substantially toward null (Firth HR 0.78 $\rightarrow$ Ridge HR 0.39), suggesting the OS estimate is sensitive to regularization and should be treated as imprecise. The $\text{TMB/BRAF} \times \text{Rx}$ OS estimate shows minimal shrinkage (0.93 $\rightarrow$ 0.87), consistent with a near-null interaction. For PFS, the $\text{CRP} \times \text{Rx}$ estimate is essentially unchanged by Ridge (Firth 0.33 $\rightarrow$ Ridge 0.32), indicating that the directional PFS finding is robust to regularization — the data support it even when the model is penalized for large coefficients. $\text{TMB/BRAF} \times \text{Rx}$ PFS shows moderate shrinkage (0.60 $\rightarrow$ 0.45).

8.4 TLR as a post-treatment response

TLR is strongly associated with both survival outcomes (OS HR 0.35, 95% CI 0.20–0.61; PFS HR 0.17, 0.09–0.32) in unadjusted analysis. Descriptively, however, a noteworthy imbalance exists: TLR-positive status is more common in the control arm (76%) than the experimental arm (53%). This is surprising if TLR captures treatment response (one would expect higher response rates with immunotherapy, or at least parity). Possible explanations include small-sample randomization imbalance, a timing artefact (TLR measured at a fixed CT which may occur during different treatment phases across arms), or genuine heterogeneity in the patient mix. The imbalance reinforces the caution against interpreting the strong TLR-survival association

as evidence of a TLR-based treatment effect: much of the TLR survival difference could be confounded by unmeasured baseline prognosis. A formal causal mediation analysis would be needed to separate the direct treatment effect from the indirect path through TLR.

9 Summary

Study: METIMMOX-1 biomarker subgroup analysis. N = 65 complete cases, 56 OS events, 48 PFS events.

Methods: Unified Firth-corrected Cox model motivated by the causal DAG: $\text{Surv} \sim \text{Age} + \text{sex} + \text{Rx} + \text{CRP} + \text{TMB/BRAF} + \text{CRP}:\text{Rx} + \text{TMB/BRAF}:\text{Rx}$. TLR excluded as a post-randomization mediator ($T \rightarrow \text{TLR} \rightarrow \text{PFS}$). Profile likelihood CIs, PLRT for interaction testing. Ridge regression (L2 penalty on main effects) as sensitivity analysis.

Proportional hazards: No violations in OS (global $p = 0.40$). In PFS, the TMB/BRAF main effect ($p = 0.021$) and TMB/BRAF $\times$ Rx interaction ($p = 0.045$) violate the PH assumption; PFS TMB/BRAF estimates should be interpreted with caution.

Interaction estimates:

Interaction	OS HR (95% CI)	PLRT p	PFS HR (95% CI)	PLRT p
CRP $\times$ Rx	0.78 (0.21–3.61)	0.73	0.33 (0.07–2.15)	0.22
TMB/BRAF $\times$ Rx	0.93 (0.30–2.85)	0.89	0.60 (0.16–2.28)	—

No interaction is statistically significant. CRP $\times$ Rx PFS (HR 0.33) is the most directionally consistent finding and is robust to Ridge regularization (Ridge HR 0.32). TMB/BRAF OS shows near-null interaction (Ridge HR 0.87 Firth); TMB/BRAF PFS violated PH.

TLR (descriptive): OS HR 0.35 (0.20–0.61); PFS HR 0.17 (0.09–0.32). TLR-positive prevalence: control 76%, experimental 53% — a directional imbalance that complicates causal interpretation. TLR is a post-treatment mediator and requires causal mediation analysis for valid inference.

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